Analytical Chemistry Cumulative Exam
Sept. 26, 2008
(Feature Article from *Analytical Chemistry*, August 1, 2005, "When Voltammetry Reaches Nanoseconds")

1. **(5 points)** Who is Christian Amatore? Why is he important to this article? What do you know about him?

2. **(10 points)** Feldberg and Smalley introduced a "temperature jump" method. What is this method for? In very general terms, how does it work? What does it avoid?

3. **(15 points)** Describe two reasons why ohmic drop destroys electrochemical information. Why do ultramicroelectrodes offer an improvement over millielectrodes with respect to this topic?

4. **(10 points)** Stray capacitances between the electrode and the solution and within the circuit represent a significant fraction of the global electrode capacitance. Thus, improvements offered by smaller and smaller electrodes with radii of $\leq 1\ \mu m$, are not as good as would be predicted. Therefore, in order for the nanosecond realm to be reached, additional strategies, not just smaller electrodes, must account for the ohmic drop. Describe one of these strategies. (Two were described in the article, one offered by Wightman and Saveant and the other by Wipf.)

5. **(20 points)** Until recently, the measurement of fast rates of electron transfers occurring within molecules was performed only through fast photophysical homogenous kinetics that generally resorted to very specific rigid tethers and photochemically excited redox centers. An easier way to measure these rates is to anchor a molecule that has a redox center on one end to the surface of an electrode, as in self-assembled molecules. Explain the theory that predicts how the first order heterogeneous rate constant for electron transfer ($\mathbf{k_{ET}}$) should drop as the distance increases that separates the two coupled centers within the transition state (long range electron transfer).

6. **(20 points)** What does a cyclic voltammogram that exhibits typical thin-layer behavior of the oxidation and reduction of a monolayer of kinetically-reversible redox species tethered to an electrode surface look like (at moderately low scan rates)? Draw the voltammogram that you might observe. What happens to the voltammogram when the scan rate, v, is increased so that $k_{ET} \times RT/(Fv) << 1$. Show a diagram of what you might expect and explain in writing what happens.

7. **(20 points)** Two examples were given of the use of ultrafast electrochemistry to study fast physical chemistry events. One involves chronoamperometry and the life of an acoustic bubble. The other involves the monitoring of charge propagation using ultrafast cyclic voltammetry within a nanoscale molecular object (polyamidoamine dendrimer functionalized by 64 $Ru(tpy)_2$ complexes, tpy = terpyridine). Pick one of these examples and describe the key conclusions determined for that system.